

Igor Garcia

Software Engineer | EU Citizen | +55(75)999823805 | igonildo7@gmail.com

LINKS

Personal website, [Github](#) , [LinkedIn](#) , [Artstation](#)

Software Engineer specializing in Unreal Engine (C++/Blueprint) and Unity Engine(C#) with 4+ years building high-performance VR/XR and game experiences. Deliver interactive simulations for automotive design validation, optimize rendering and UX across Vive and Meta Quest devices, and create polished gameplay and UI for multiplayer titles. Cross-engine fluency in Unity (C#) and Godot , including AI-enabled interactive apps integrating Azure Cognitive Services and OpenAI . Proficient in Git/Perforce version control and CI/CD build automation (GitHub Actions). Complementary strengths in the 3D pipeline (ZBrush , Blender , Maya , CATIA), with a focus on performance, collaboration, and rapid iteration.

SKILLS

Game Development:	Unreal Engine, Unity, Godot, C++, C#, Python, Git
Web Development:	React, JavaScript, HTML, CSS, Tailwind
3D modeling:	Blender, Substance Painter 3D, Maya

WORK EXPERIENCE

Software engineer/C++ developer, Ford Motor Company

Jan 2023 — Present

- Engineered immersive VR validation tools in Unreal Engine 5 with automated CI/CD build pipelines (GitHub Actions), replacing physical prototypes and reducing material costs by approximately 40% annually .
- Optimized high-fidelity visualizations for **5+ VR platforms** (HTC Vive, Meta Quest Pro), achieving stable **60 FPS** performance even with high-poly automotive assets (**5M+ polygons**).
- Accelerated design iteration cycles by **50%** , allowing stakeholders to validate vehicle features in real-time rather than waiting weeks for physical models.
- Built real-time interactive simulations utilizing C++ and Blueprints, ensuring immediate feedback loops for design validation workflows.

Software engineer/C#, Cafundo Creative Studio

Jun 2025 — Present

- Architected an AI-powered interactive totem in Unity (C#) for high-traffic public environments, serving **over 500 daily interactions** with zero downtime.
- Seamlessly integrated Azure Cognitive Services and OpenAI APIs, reducing voice-to-response latency by **30%** to create a natural conversational flow.
- Designed intuitive UI/UX systems focused on accessibility and seamless real-time user interaction.

Unreal Engine 5 Developer, Blue Gravity Studios

Jan 2024 — Sep 2024

- Developed core gameplay systems using C++ and Blueprints, implementing automated testing for core systems and improving code modularity and reducing bug-fixing overhead by 20% for the development team.
- Engineered network replication logic for the multiplayer title 'SkateNationXL', optimizing bandwidth usage by **25%** to support smooth online sessions.
- Implemented a scalable UI system for both keyboard and gamepad, reducing input lag by **15ms** and ensuring 100% compatibility across PC and console builds.
- Collaborated with multidisciplinary teams to optimize draw calls and texture memory, enhancing overall game performance on lower-end hardware.

Virtual Reality Researcher, Ford Motor Company

Aug 2021 — Dec 2022

- Spearheaded R&D for automotive VR applications, establishing foundational workflows that were later adopted as the **company standard for virtual prototyping** .
- Collaborated with Design and Engineering teams to translate physical requirements into virtual specifications, resulting in a **20% improvement** in user assessment accuracy.
- Developed initial prototypes for virtual user experience testing, validating key interaction concepts for future production.

PROJECTS

Professional Measurement Tool | Unreal Engine C++ (Fab / Unreal Marketplace, 2026) 2026

Author of a C++ editor plugin that measures distances and enclosed polygon areas in-editor using USplineComponent, with real-time visual feedback and a dedicated Editor Mode.

- Decoupled architecture using UInterface and multicast delegates, separating actor, widget, snap, and label logic into independent components.
- Three snap modes (Off / Ground / Surface) with six-axis cardinal traces for precise point placement on complex geometry.
- Custom Editor Mode (FEdMode) with Slate toolkit panel for one-click spawning, selection, and property editing.

[Project link on FAB: <https://www.fab.com/listings/5c2a20d1-1537-4ca3-9255-405d38b79b05>]

Runtime Skinned Mesh Exporter (Fab / Unreal Marketplace, 2025) 2025

Author of a C++/Blueprint plugin that exports skinned Skeletal & Static Meshes at runtime to an OBJ + MTL + PNG package.

- Fully asynchronous pipeline: GPU→CPU transfer through FRHIGPUTextureReadback and PNG compression on the thread-pool, ensuring hitch-free operation.
 - Compatible with UE 4.27 → 5.6 via version-conditional code;
 - Managed the entire product lifecycle: ideation, development, documentation and multiproject testing
- [Project link on FAB: [Runtime Skeletal \(Skinned\) / Static Mesh Exporter | Fab](#)]

Real-Time Object Obstruction Detector | Unreal Engine C++ 2025

Built a high-performance C++ computer vision algorithm to compute real-time occlusion percentages between virtual objects in Unreal Engine.

- Replaced legacy offline analysis with a custom runtime solution, reducing validation time from ~2 hours to seconds.
- Maintains precise spatial measurements while preserving engine performance.

Unused Plugins Manager | Unreal Engine C++ 2025

Built an Unreal Engine C++ tool that scans projects for unused plugins and produces a removable list.

- One-click removal of redundant modules reduces final package sizes and build times.
- Automates previously manual auditing, promoting leaner architecture and faster deployment.

LANGUAGES

- **Languages:** English (C2), Portuguese (Native)

EDUCATION

University of Feira de Santana (UEFS), Feira de Santana

Computer engineering degree